Student Name: Cuong Dang
Course: ITAI-2372-Artificial Intel Applications
Date: June 12, 2026

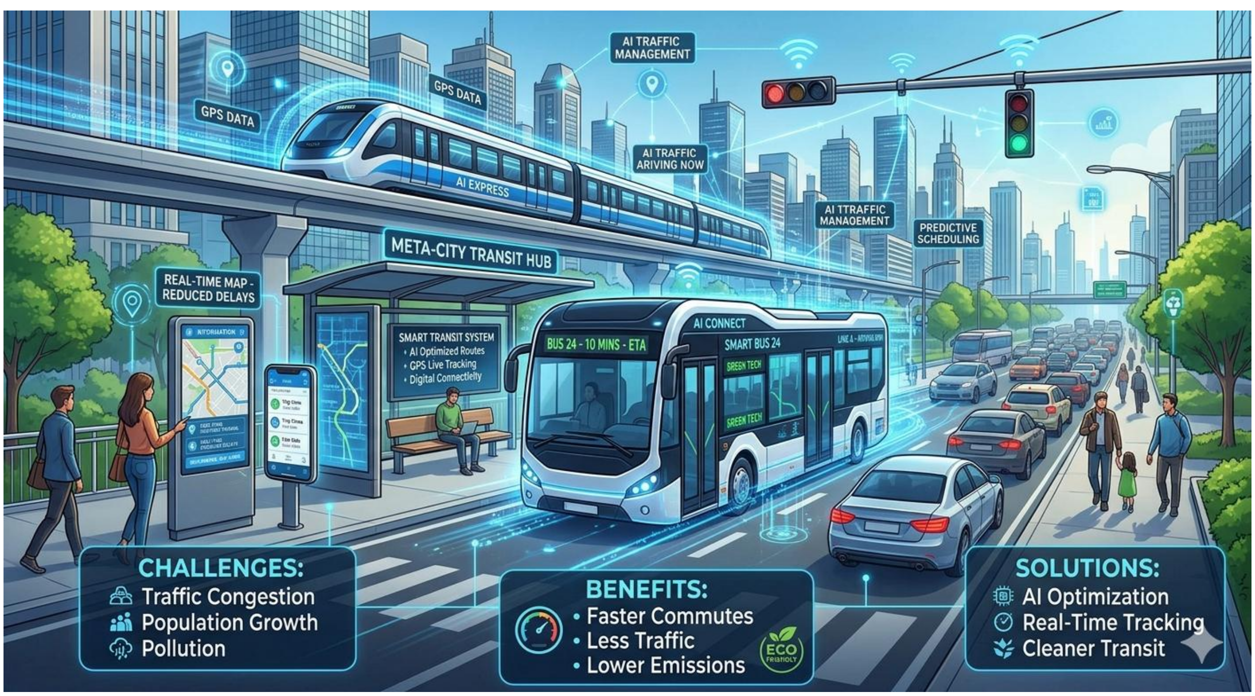
A05 Module 05 Smart Cities Module Reflections

Smart Public Transportation industry

1. Industry Overview

The industry I chose for this reflection is the **Smart Public Transportation industry**. This industry focuses on using digital technology, GPS tracking, and Artificial Intelligence (AI) to manage and improve public transit systems like buses, trains, and subways. It serves daily commuters, city travelers, and local transit authorities.

This industry is highly relevant today because big cities are facing massive population growth. This growth causes terrible traffic congestion, longer commute times, and heavy air pollution from cars. Traditional transit systems with fixed schedules cannot handle the high number of passengers anymore. By using AI, this industry makes public transit faster, more reliable, and cleaner, encouraging more people to leave their cars at home.



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2. Current AI Trends

AI is actively being used right now to solve major public transit problems. Based on the case studies, here are two major ways AI is deployed in this field:

- **Real-Time Demand and Schedule Optimization:** In **Vista City**, the transit authority uses AI algorithms to track bus locations and passenger demand in real time. Instead of buses running on rigid, old schedules, the AI automatically adjusts bus frequencies based on how many people are waiting at stations. This reduces waiting times and stops buses from running completely empty.
- **Traffic Flow and Congestion Reduction:** In **Metroville**, the city deployed an AI platform that connects traffic cameras and transit data. The system uses machine learning to dynamically change traffic light timings when buses or emergency vehicles are approaching. This trend successfully reduced overall traffic congestion by 30% and improved emergency response times by 40%.

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3. Ethical Implications

While AI makes public transportation highly efficient, it introduces serious ethical challenges for society. Two major concerns are:

- **Loss of Privacy and Constant Surveillance:** To track passenger demand and optimize routes, smart transit systems use cameras with facial recognition and track mobile phone signals at stations. This means citizens are under constant surveillance just to take a bus or train. If hackers breach these transit databases, the daily travel histories and personal habits of millions of citizens could be stolen.
- **Job Displacement for Transit Workers:** As AI takes over route planning, scheduling, and eventually driving, many human workers face the risk of losing their jobs. In the future, autonomous buses and trains will not need human drivers. Ticket booth operators and manual schedulers are already being replaced by AI systems, which create financial hardships for these workers and their families.

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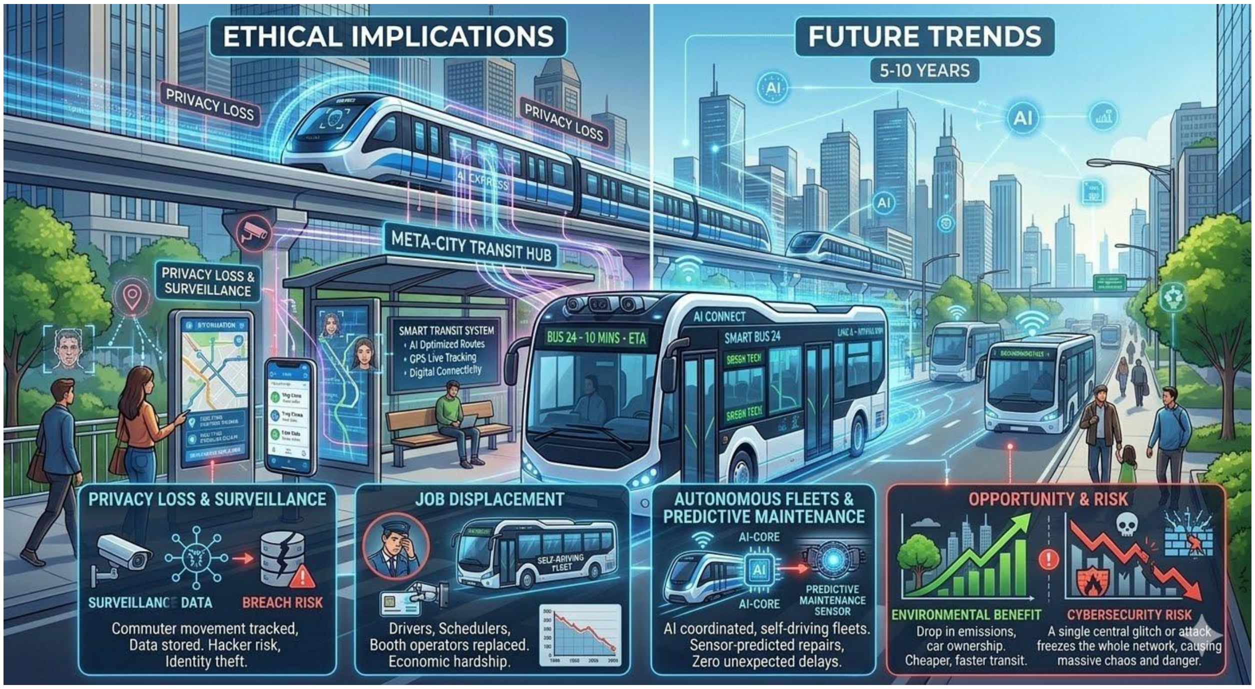

4. Future Trends

In the next 5 to 10 years, the Smart Public Transportation industry will look completely automated and interconnected. A major innovation on the horizon is the use of **Autonomous Flying Taxis (eVTOL)** to bypass ground traffic completely. AI will manage these flying networks to prevent collisions and schedule rapid flights across the city sky. Furthermore, to make these flying vehicles truly eco-friendly, they will need advanced clean energy. A sustainable solution for this is **hydrogen fuel technology**. By using hydrogen, flying taxis can achieve long flight ranges and quick refueling times with zero toxic emissions, completely transforming green urban mobility.

- **The Opportunity:** Cities will experience a massive drop in carbon emissions. Public transit will become so fast and convenient that private car ownership will drop significantly, helping the environment.
- **The Risk:** A total reliance on software creates huge cybersecurity risks. If a single central AI system manages ground buses and flying taxis all at once, a software bug or a cyberattack could freeze the entire city's transportation network instantly.

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5. Your Reflection

What surprised me most is how much AI can improve transit metrics. Seeing that Metroville cut traffic congestion by 30% just by using smart AI algorithms made me realize that technology can fix real-world frustrations that we face every day.

I would like to work in the Smart Public Transportation industry. In fact, I am very passionate about this field and have already built a **GitHub project focusing on hydrogen fuel systems for modern transportation vehicles**.

(Link GitHub here: https://github.com/binxixi23/Green-Hydrogen-Revolution).

It is exciting to develop technology that directly helps millions of people travel safely and quickly. Even though the risks like cybersecurity are tough, working in this field would allow me to apply my software skills and clean energy knowledge to build a greener, faster future for our cities.